

Wasserstein Gradient Flows

Exercise statements

This version contains the exercises only. Hints and worked solutions are available in the companion PDF.

Exercise 1 (φ -divergences and Wasserstein gradient flow).

Let $\nu \in \mathcal{P}_2(\mathbb{R}^d)$ be a reference probability measure and let $\varphi : [0, \infty) \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex, lower semicontinuous and C^2 on $(0, \infty)$ with $\varphi(1) = 0$. For a probability measure α such that $\alpha \ll \nu$ with density $r := \frac{d\alpha}{d\nu}$, the φ -divergence of α with respect to ν is defined by

$$D_\varphi(\alpha \mid \nu) := \int_{\mathbb{R}^d} \varphi\left(\frac{d\alpha}{d\nu}(x)\right) d\nu(x) = \int_{\mathbb{R}^d} \varphi(r(x)) d\nu(x).$$

- (1) Show that $D_\varphi(\alpha \mid \nu) \geq 0$.
 - (2) Consider the functional $f(\alpha) := D_\varphi(\alpha \mid \nu)$ on $\mathcal{P}_2(\mathbb{R}^d)$. Assume that both ν and α admit smooth strictly positive densities with respect to Lebesgue measure.
 - (a) Compute the first variation $\frac{\delta f}{\delta \alpha}(\alpha)(x)$ and deduce the Wasserstein gradient $\mathbb{W}f(\alpha)(x)$.
 - (b) Write the corresponding Wasserstein gradient flow PDE.
 - (c) Specialize your formulas to the case $\varphi(s) = s \log s$.
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Exercise 2 (Gradient flows of Dirichlet energy and Fisher information).

To avoid boundary terms, work on the flat unit-volume torus $\mathbb{T}^d$, with a smooth strictly positive density ρ and $d\alpha = \rho dx$. The same local differential expressions hold on $\mathbb{R}^d$ when the required integrations by parts are justified.

- (1) For the Dirichlet energy $\mathcal{E}(\alpha) = \int |\nabla \rho|^2 dx$:
 - (a) Compute the first variation and the unconstrained $L^2(dx)$ descent equation $\partial_t \rho = -\delta \mathcal{E} / \delta \rho$.
 - (b) Compute the Wasserstein gradient and its continuity-equation PDE.
- (2) Repeat these computations for the Fisher information

$$\mathcal{I}(\alpha) = \int \rho |\nabla \log \rho|^2 dx = \int \frac{|\nabla \rho|^2}{\rho} dx.$$

Does its unconstrained L^2 flow preserve mass? How should it be modified to preserve $\int \rho = 1$? All evolution equations are to be understood while the solution is smooth and positive.

Exercise 3 (Gaussian evolution under linear dynamics).

Let $(X^{(\ell)})_{\ell \in \mathbb{N}}$ be a sequence of random variables in $\mathbb{R}^d$ defined by the linear recursion

$$X^{(\ell+1)} = X^{(\ell)} - \tau A^{(\ell)} X^{(\ell)}, \quad \ell = 0, 1, 2, \dots,$$

where $\tau > 0$ is a fixed step size and, for each ℓ , $A^{(\ell)} \in \mathbb{R}^{d \times d}$ is a deterministic matrix. Denote by $\alpha^{(\ell)}$ the law of $X^{(\ell)}$.

- (1) Assume that the initial condition is Gaussian $X^{(0)} \sim \mathcal{N}(0, \Sigma^{(0)})$, for some symmetric positive semidefinite covariance matrix $\Sigma^{(0)} \in \mathbb{R}^{d \times d}$. Show that each $X^{(\ell)}$ is also Gaussian, and describe the evolution of the law $\alpha^{(\ell)} = \text{Law}(X^{(\ell)})$ by giving an explicit recursion for the covariance matrices $\Sigma^{(\ell)}$.

- (2) Consider now the continuous time ODE $\dot{X}_t = -A_t X_t$, $t \geq 0$, with $A_t \in \mathbb{R}^{d \times d}$ a (say, continuous) time dependent matrix and random initial condition $X_0 \sim \mathcal{N}(0, \Sigma_0)$. Explain why X_t is Gaussian for every $t \geq 0$, say $X_t \sim \mathcal{N}(0, \Sigma_t)$, and, by considering the limit $\tau \rightarrow 0$ in the discrete time recursion from part (1), show that $(\Sigma_t)_{t \geq 0}$ solves a matrix ODE that you should identify.

Exercise 4 (Wasserstein flow for quadratic kernels).

Let $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ and denote its mean and covariance by

$$m(\alpha) := \int_{\mathbb{R}^d} x d\alpha(x), \quad \Sigma(\alpha) := \int_{\mathbb{R}^d} (x - m(\alpha))(x - m(\alpha))^\top d\alpha(x).$$

Consider the functionals

$$f(\alpha) := \iint_{\mathbb{R}^d \times \mathbb{R}^d} k(x, y) d\alpha(x) d\alpha(y)$$

on the Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$, where the kernels k are given below.

- (1) Let $k(x, y) = \langle Ax, y \rangle$, $A \in \mathbb{R}^{d \times d}$.
- (a) Express $f(\alpha)$ in terms of $m(\alpha)$ and A .
 - (b) Compute the Wasserstein gradient $\nabla f(\alpha)$.
 - (c) Write the Wasserstein gradient flow PDE, and, for a general initial condition $\alpha_{t=0} = \alpha_0$, give an explicit expression of α_t in terms of α_0 .
- (2) Let $k(x, y) = \langle Ax, y \rangle \langle Bx, y \rangle$, $A, B \in \mathbb{R}^{d \times d}$.
- (a) Compute the Wasserstein gradient $\nabla f(\alpha)$.
 - (b) Using the previous exercise on linear ODEs with Gaussian initial data, show that if $\alpha_{t=0} = \mathcal{N}(0, \Sigma_0)$, then $\alpha_t = \mathcal{N}(0, \Sigma_t)$ for as long as the covariance ODE exists, and derive the matrix ODE satisfied by $(\Sigma_t)_{t \geq 0}$.

Further exercises

Exercise 5 (Mean-field two-layer linear network under Wasserstein flow).

Let $\rho \in \mathcal{P}_2(\mathbb{R}^d)$ be the data distribution and let $h \in L^2(\rho)$ be a target function. For a parameter distribution

$$\alpha \in \mathcal{P}_2(\mathbb{R} \times \mathbb{R}^d), \quad x = (u, v) \in \mathbb{R} \times \mathbb{R}^d,$$

consider the two-layer linear mean-field model

$$g_\alpha(z) := \int u \langle z, v \rangle d\alpha(u, v), \quad z \in \mathbb{R}^d,$$

and the quadratic loss

$$f(\alpha) := \int_{\mathbb{R}^d} |g_\alpha(z) - h(z)|^2 d\rho(z).$$

- (1) Show that f can be written as a quadratic functional of α of the form

$$f(\alpha) = \iint k(x, y) d\alpha(x) d\alpha(y) + \int b(x) d\alpha(x) + c,$$

where $x = (u, v)$, $y = (u', v')$, and identify explicit formulas for $k(x, y)$ and $b(x)$ in terms of ρ and h . You may leave the constant c implicit.

(2) Consider the Wasserstein gradient flow of f on $\mathcal{P}_2(\mathbb{R} \times \mathbb{R}^d)$:

$$\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0, \quad v_t(x) = -\mathbb{W}f(\alpha_t)(x).$$

Using the structure from part (1) and the previous exercise on quadratic kernels, show that if

$$\alpha_{t=0} = \mathcal{N}(0, \Sigma_0),$$

then α_t remains Gaussian for all $t \geq 0$.

Exercise 6 (Wasserstein flow of W_2^2).

Fix $\beta \in \mathcal{P}_2(\mathbb{R}^d)$. For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ define $f(\alpha) := W_2(\alpha, \beta)^2$. For the first-variation calculation, assume α has a smooth strictly positive density and that the usual envelope differentiation is justified for the perturbations considered. For the flow take an absolutely continuous initial law $\alpha_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Consider the quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$.

- (1) Using the Kantorovich dual problem, and using the envelope theorem (that was used the same way in the previous exercise sheet for discrete measure), compute the first variation $\delta f(\alpha)$.
- (2) Assume α is absolutely continuous. Leveraging Brenier's theorem connecting the transport map T to the dual potential, compute the Wasserstein gradient $\mathbb{W}f(\alpha)$.
- (3) Relate the Wasserstein flow $(\alpha_t)_{t \geq 0}$ to the McCann interpolation (you should use Benamou-Brenier dynamical formulation which details a valid field advecting the interpolant).